POST MIDSEM

Goods Market

$$C = C_0 + c_1(Y - T)$$
$$I = b_0 + b_1 Y - b_2 i$$
$$Z = C + I + G$$
$$i = \bar{i}$$

Money Market

$$\frac{M}{P} = d_1 Y - d_2 i$$

① Solve eq levels of Y, i

② " " of real money supply

③ Show that an ↑ in $\underbrace{\text{public}}$ saving leads to ↓ in I

public → BS ↓ Budget Surplus

Ans. $Y^* = Z$

① $$Y^* = C_0 + c_1(Y - T) + b_0 + b_1 Y^* - b_2 \bar{i} + G$$

$$Y^* = \frac{C_0 - c_1 T + b_0 - b_2 \bar{i} + G}{1 - c_1 - b_1}$$

$$I^* = b_0 + b_1\left(\frac{C_0 - c_1 T + b_0 - b_2 \bar{i} + G}{1 - c_1 - b_1}\right) - b_2 \bar{i}$$

Cut down on budget deficit → Public saving increasing

$G \downarrow \Rightarrow (T - G) \uparrow$

$$I = b_0 + b_1\left(\frac{C_0 - c_1 T + b_0 - b_2 \bar{i} + G}{1 - c_1 - b_1}\right) - b_2 \bar{i}$$

$$\frac{\partial I}{\partial G} = \frac{b_1}{1 - c_1 - b_1} > 0 \quad (\text{Assuming } b_1 + c_1 < 1)$$

Deficit going down ⇒ Investment ↓

Private savings → Finance BS

Private savings → Investments

I = Pvt saving + Public saving
$= S + (T - G)$

Suppose fixed

$T\uparrow, G\downarrow, (T-G)\uparrow \longrightarrow$ Investment should've increased but S is not fixed in reality

① Taxes can affect
② $\Delta G \rightarrow \Delta Y = \frac{\Delta G}{1-c}$

Fall in private savings is greater than increase in public savings

OPEN ECONOMY IS-LM MODEL

Domestic output

Exports
Imports

$A = f(Y, i) \longleftarrow A = C + I + G$

Spending by domestic residents
$\neq$
Spending on domestic output

Open economy

$= C + I + G + \underbrace{(X - M)}_{\text{export - import}}$

Trade balance

Net exports = exports − imports

$X(Y_f, R)$ $\quad Q(Y, R)$

Y_f : Income of foreigners
R: Real exchange rate

Eg:- Equate \$ and ₹ it takes to buy 1kg of rice

$R = \frac{eP_f}{P}$; If purchasing power is in parity $R=1$

$\implies R > 1$, foreign currency is stronger
Price in domestic country goes up as demand shifts and $R \rightarrow 1$

$$NX = f(Y_f, Y, R)$$

① $Y_f \uparrow$, Demand for our goods would increase $\Rightarrow X \uparrow \Rightarrow NX \uparrow$

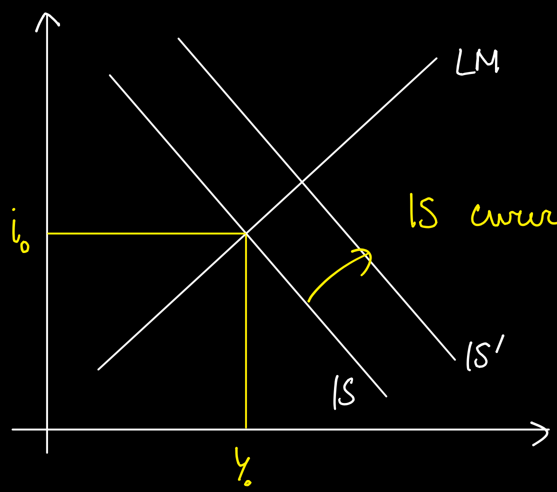
② $Y \uparrow \Longrightarrow M \uparrow \Rightarrow NX \downarrow$

③ $R \uparrow \Longrightarrow$ Domestic goods get cheaper so greater demand for our goods $\Rightarrow X \uparrow \Rightarrow NX \uparrow$

MARGINA PROPENSITY TO IMPORT

$$NX = NX(Y, Y_f, R)$$

IS curve $\quad Y = A(Y, i) + NX(Y, Y_f, R)$



IS curve shifts due to change in NX

	$\uparrow$ in home expenditure	$\uparrow$ in foreign expenditure	real (domestic currency) depreciation
Income	+	+	+
Net exports	−	+	+

Cases

I Say $G\uparrow \Rightarrow Y\uparrow \longrightarrow$ Demand for imports $\uparrow \Longrightarrow Y_f\uparrow$

II $\underbrace{\text{Depreciation of domestic currency}}_{\text{Only benefits domestic country}} \Longrightarrow$ Exports $\uparrow \Rightarrow$ Domestic output $\uparrow$

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Current A/C + Capital A/C = 0

Account

Pvt sector

Official reserve transactions $\rightarrow$ of foreign currency

International Economics

Current account deficit : $\underbrace{\text{spending is greater than earning}}_{\text{country scale}}$

Balance of Payment Surplus

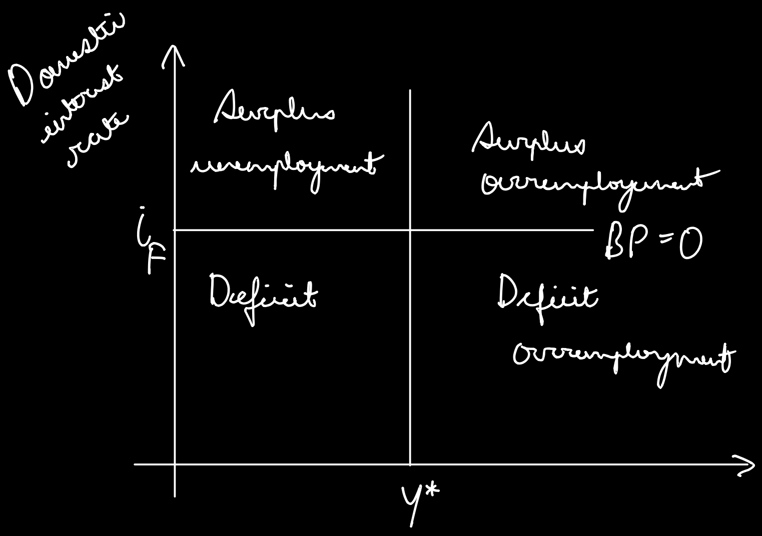
$\uparrow$ in official reserves = current A/C surplus + net pvt capital gain is low

Price of imports

World interest rate = i_f

$BOP = NX(Y, Y_f, R) + CF(i - i_f)$

Capital flow



Domestic interest rate
Surplus unemployment
Surplus overemployment
i_F
BP = 0
Deficit
Deficit overemployment
Y*

MUNDELL - FLEMING MODEL

* Fixed Exchange Rate Regime

i_w - World Interest Rate

Say central banks wants to embark on a contractionary monetary policy $\Rightarrow$ $i > i_w$

$i - i_w > 0 \Rightarrow$ Inflow of capital from foreign sources

BP surplus $\Rightarrow$ Appreciation of domestic currency

$\Downarrow$

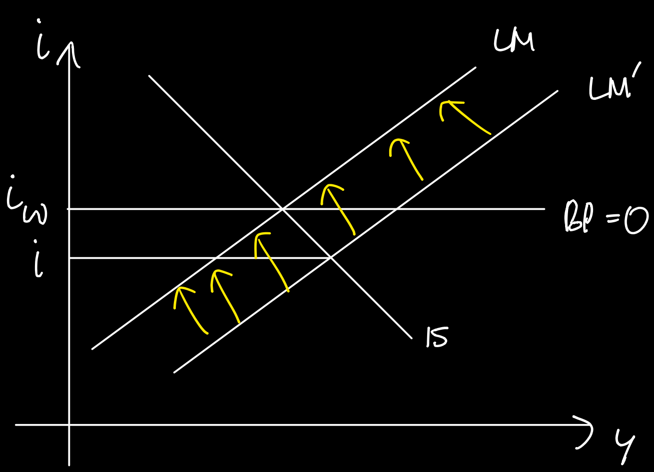
Excess demand for domestic currency

Central Bank involves in open market operations, supply of domestic currency increases $\Rightarrow$ bringing down interest rates

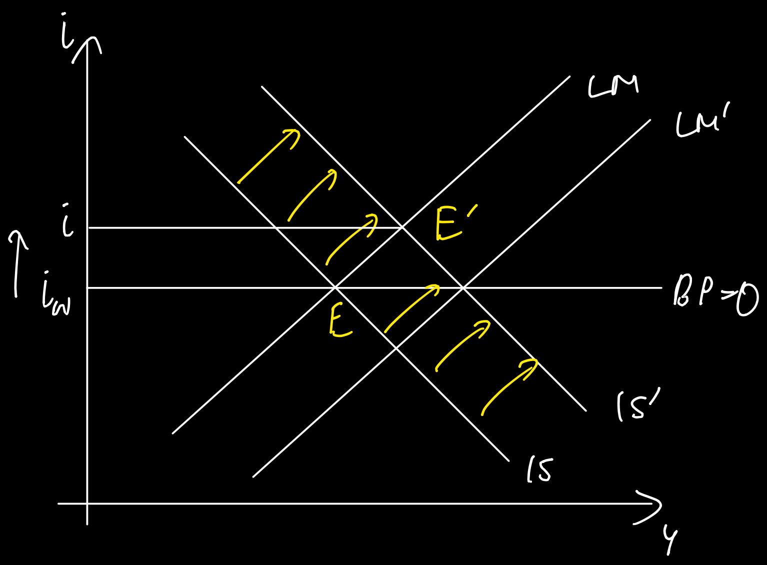
Central bank buys foreign currency from foreign exchanges which are part of the domestic economy, leading to increased supply of domestic currency.

✓ Capital perfectly mobile

✓ Fixed ER



* Monetary policies have little effect on output as interest rates are fixed at i_w



Buy foreign currency in exchange of domestic currency

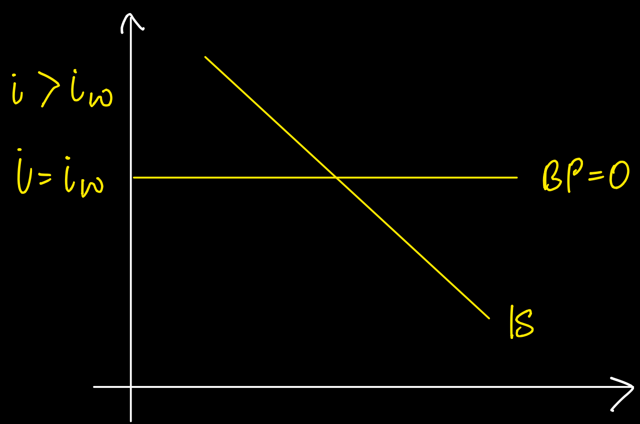
Appreciation of domestic currency.

Perfect Capital Mobility

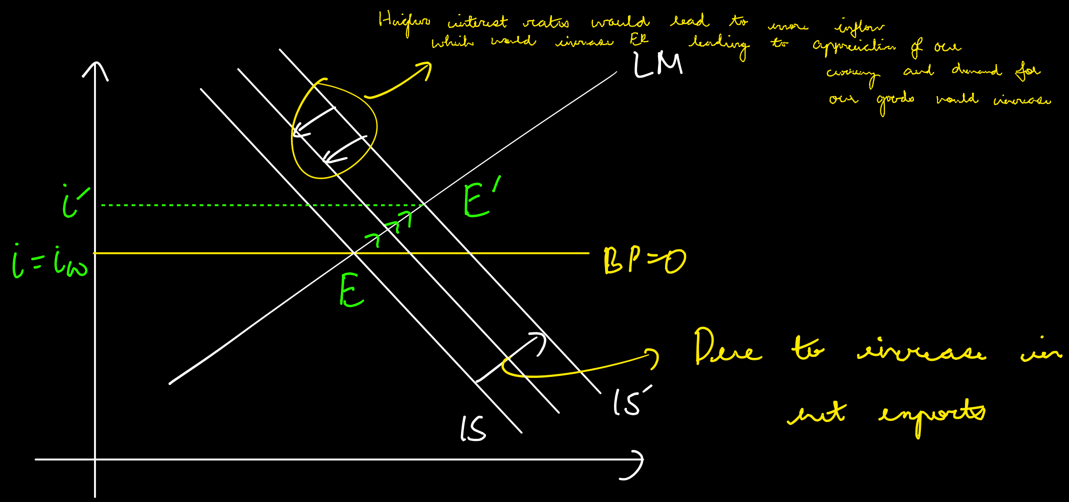
$\hookrightarrow M^S$ endogeneous $\longrightarrow$ under fixed exchange rate regime

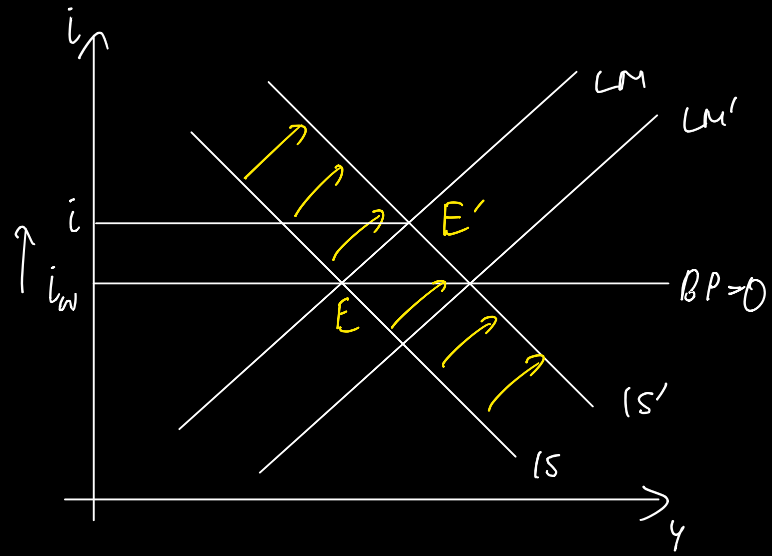
Fixed ER as central bank has an obligation to meet the new supply of money.

FLEXIBLE EXCHANGE RATE (Fiscal policy is ineffective but monetary policy is completely effective)



Appreciation of currency as investments would increase due to higher interest rates being offered.





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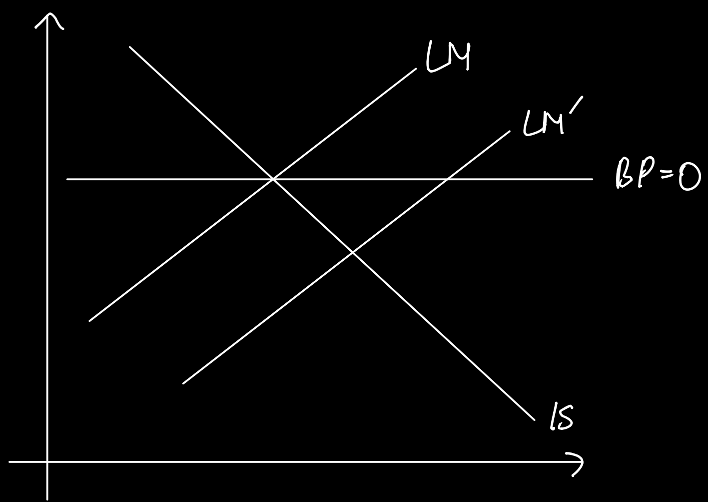

Monetary policies have a +ve impact only on the domestic economy and might have a -ve impact on rest of the world.

Fiscal policy has a +ve impact on the whole world.

Eg:-

	Fiscal Expansion	Monetary Expansion
*USA	2.7%	5.8%
Japan	0.4%	-0.6%
Germany	0.5%	-0.8%

* Expansion only performed in the USA.

BETTER-THY-NEIGHBOUR

Prosperity at the cost of a -ve impact on the trading partners (monetary expansionary policy)

Aggregate demand flows to the country which has depreciated its currency through expansionary monetary policy. Global AD stays the same.

Say both countries share the same experiences.

Country A Country B

Eg:- Countries have entered into monetary unions, say EU so that monetary decisions can be made in synchronisation

Competitive Devaluation is not a solution over the long run, as the global AD remains the same and the overall global interest rate would be same after a round of competitive devaluation by all countries. countries after a round of competitive devaluation

Solution :- Complete transparency and synchronisation of policies

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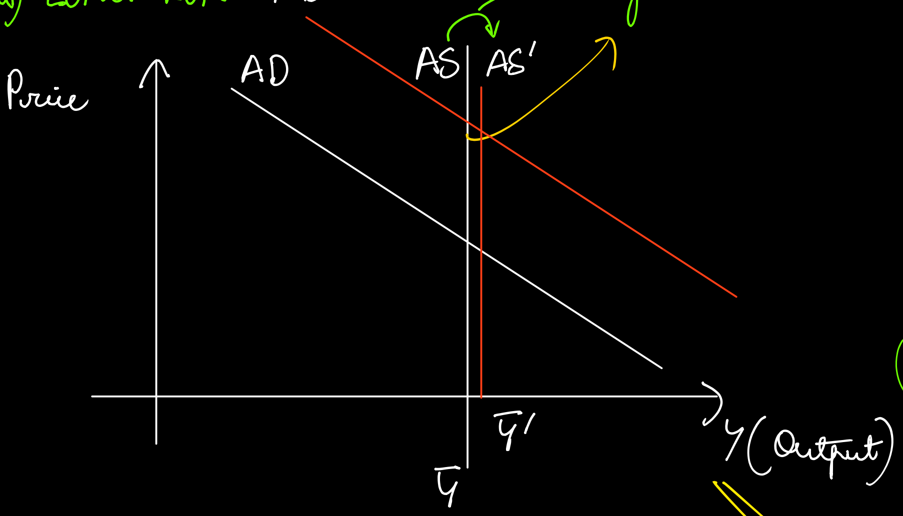
AGGREGATE SUPPLY

Classical

(1, 2-10 years) LONG RUN

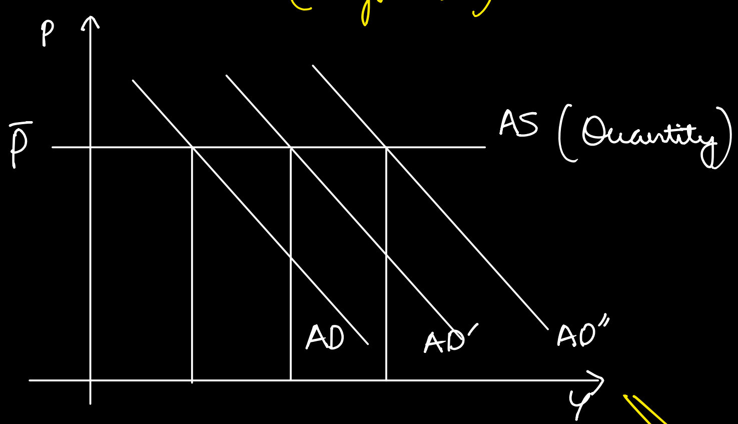
1 year



Over an episode of inflation

Long run scenario is where most resources are almost completely utilised, say skilled employment or so cannot be increased so supply cannot be increased and remains fixed over a considerable period of time.
The only option is to increase prices.

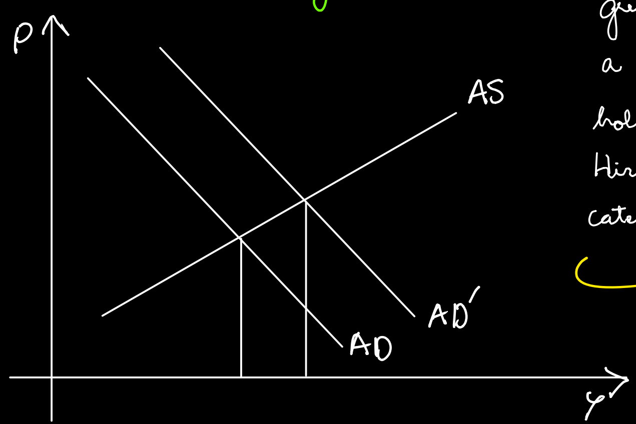
SHORT RUN (Keynesian)

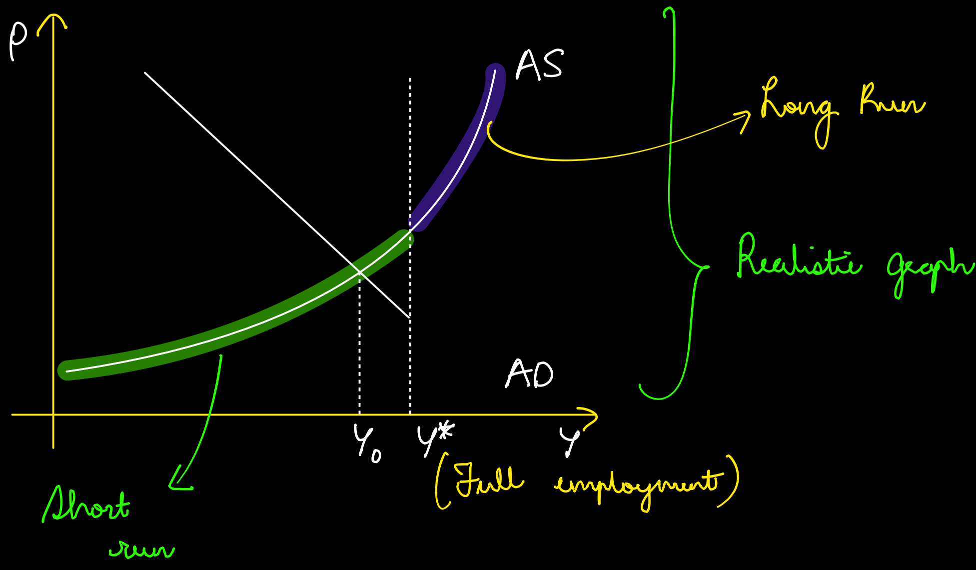


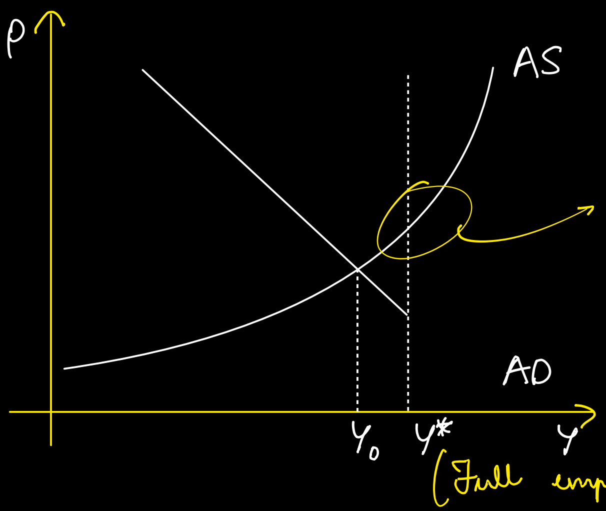
Keynes's model was developed during the great depression, increase in AD was a rarity. Even if AD increased getting hold of people to work was very easy. Hiring and firing were very easy therefore could easily cater to any potential rise in AD.

Not a realistic assumption in today's setting

MEDIUM RUN (1 year)







Tricky from policy makers POV as if demand curve shifts at this point prices could shoot up rapidly leading to inflation

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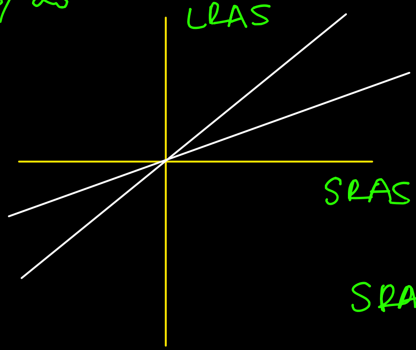
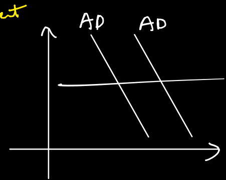


Over the short run shift in AD will have the most impact, without crowding out

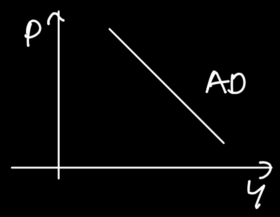


SRAS : Short run AS

$$\left(\frac{\bar{M}}{P}\right)^S$$

* If prices increase, real money supply falls and demand would fall as AS = AD

AD is a $\underbrace{\text{relation}}_{\text{inverse relation}}$ between P & Y.

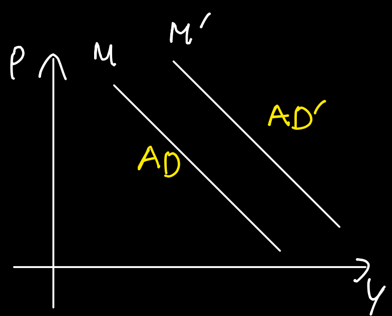


$$\boxed{M \times V = Y \times P}$$ ⟹ Quantity Theory Money

M: Money Supplied

V: Velocity of money → No. of times money turns over

If M goes up, AD curve would be shifted



M′ > M

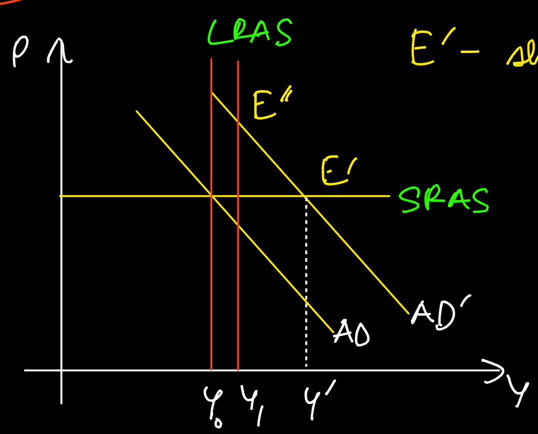
SUPPLY SIDE ECONOMICS ⟹ Productive Capacity of economy increases only by focussing on supply

Proponents argue that tax (T↓) cuts increase production as people have better incentive to work more.

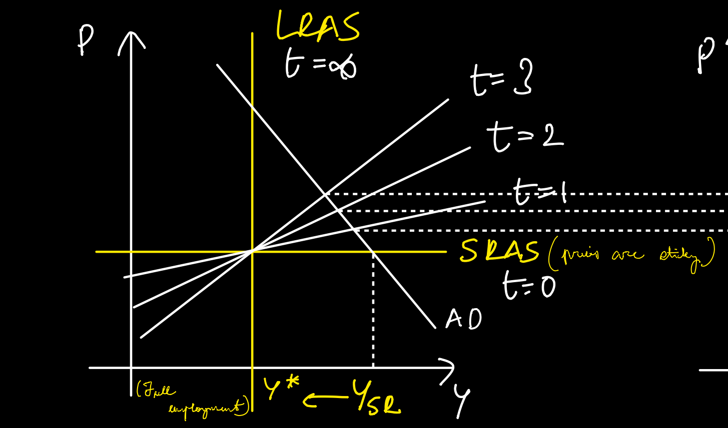
→ Boost from supply side ← Not completely true as T↓, consumption increases and prices are sticky (fixed) in the short run.

Untrue

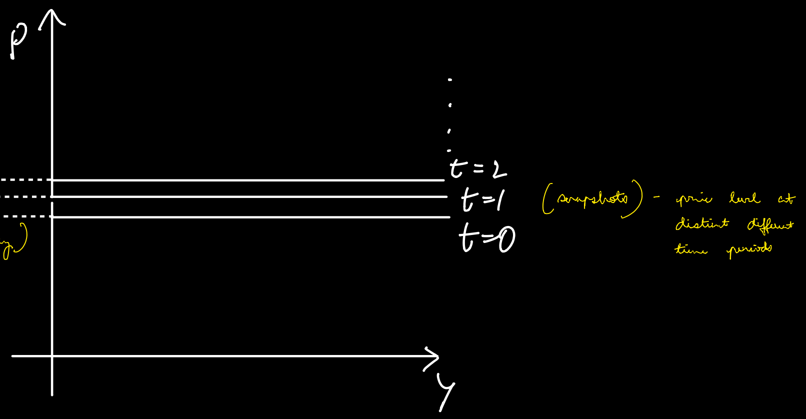
Rather production increases due to increase in AD



E′ – short run



(a)

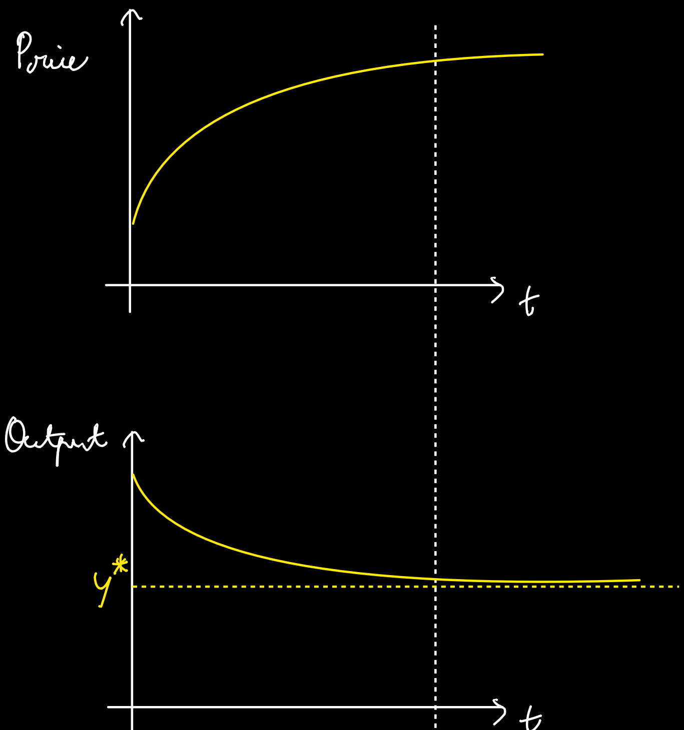


(b)

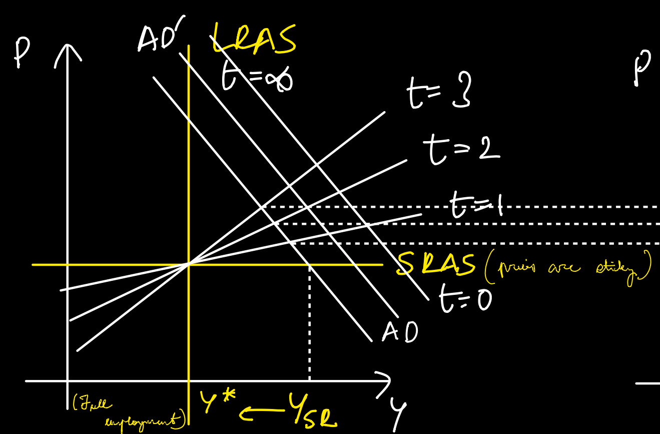
$$P_{t+1} = P_t \left[1 + \lambda \underbrace{(Y - Y^*)}_{\text{output gap}} \right]$$

$$\Rightarrow \quad P_{t+1} > P_t \quad \text{if} \quad Y > Y^*$$

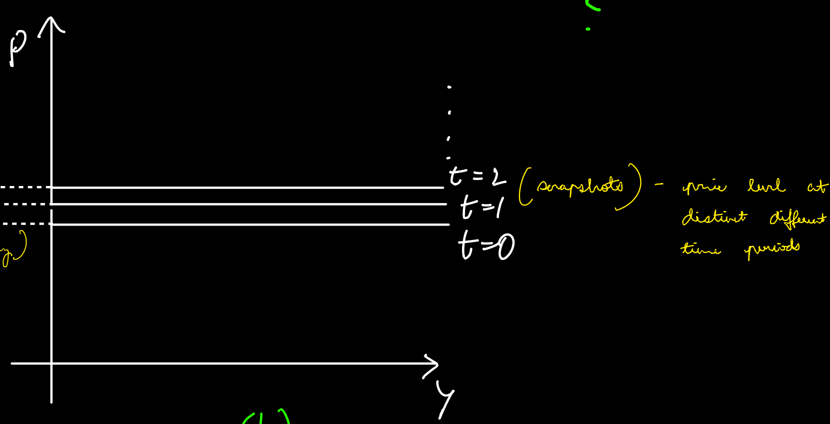
λ = speed of adjustment



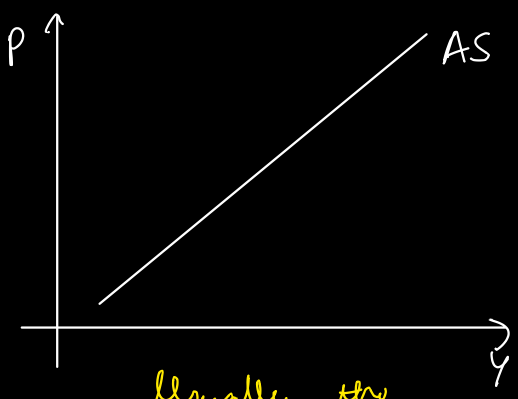
What if AD kept increasing ?



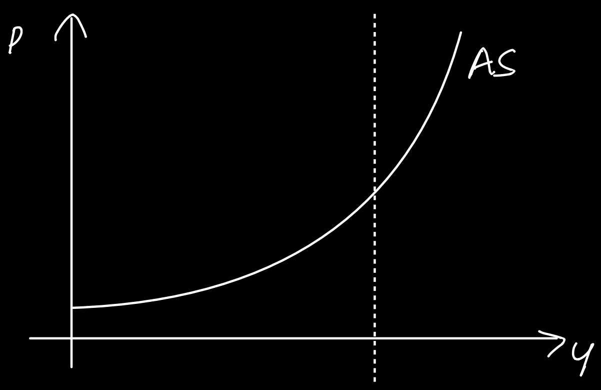
(a)



(b)



Usually the medium run



Realistic

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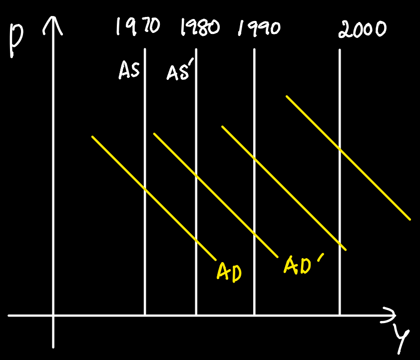
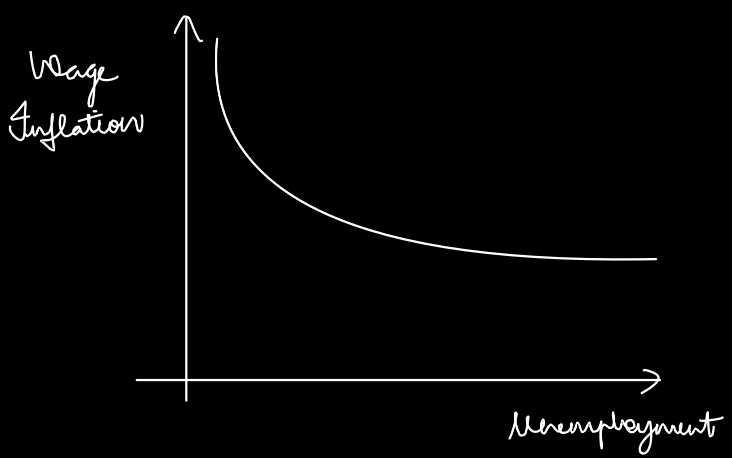


AS increased more or less at the same rate over a course of 10 years, cannot say the same about AD

Much of the reason of increase in prices over the years can be attributed to increase in AD relative to AS. } Prices over a very long period

UNEMPLOYMENT & INFLATION

→ Tradeoff ←



PHILIP'S CURVE

P → Wage inflation

Y → Unemployment

$$g_w = \frac{W_{t+1} - W_t}{W_t} \longrightarrow (A)$$

$$\boxed{g_w = -\epsilon (u - u^*)} \longrightarrow (B)$$

ϵ :- adjustment rate

g_w :- wage inflation

Wages and prices will start increasing until a new equilibrium is achieved.

Eq (A) in (B)

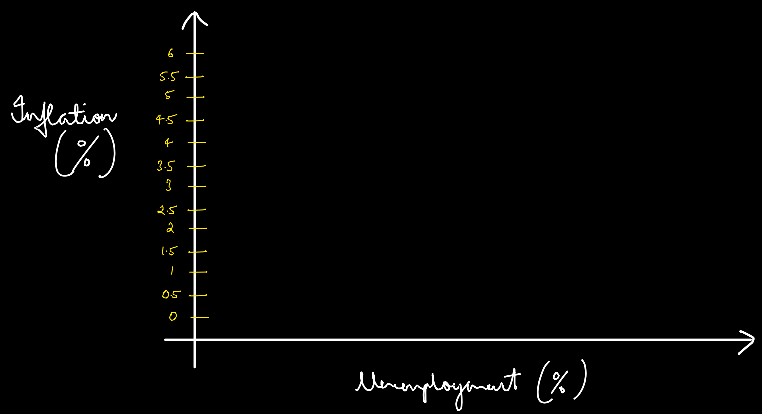
$$W_{t+1} = W_t \left(1 - \epsilon (u - u^*)\right)$$

$$u < u^* \Longrightarrow W_{t+1} > W_t$$

Wage and price inflation are essentially the same.

$$\boxed{\pi = \frac{W_{t+1}}{W_t} = 1 - \epsilon (u - u^*)}$$

Phillip's Curve → Policy Tradeoff



Eg :- $\frac{W_{t+1}}{W_t} = 0.03$, $\frac{P_{t+1}}{P_t} = 0.1$

Real wage inflation $= \frac{P_{t+1}}{P_t} - \frac{W_{t+1}}{W_t}$

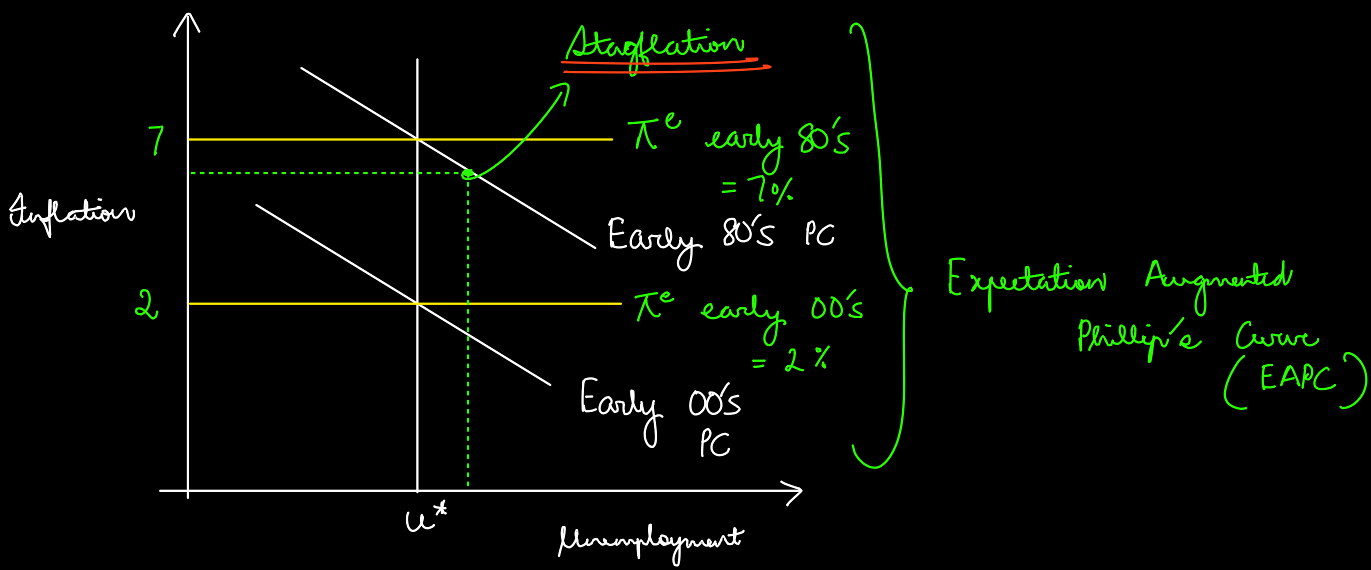
$= -0.07$

π^e :- expectated inflation $\Longrightarrow$ Used by unions use for bargaining.

$$\boxed{\pi = \pi^e - \epsilon(u - u^*)}$$

$\longrightarrow$ Modified Phillip's Curve

π :- actual inflation



STAGFLATION : Both high inflation and unemployment

Two Phillip's Curves having the same slope means same level of tradeoffs.

In classical economics prices adjust quickly.

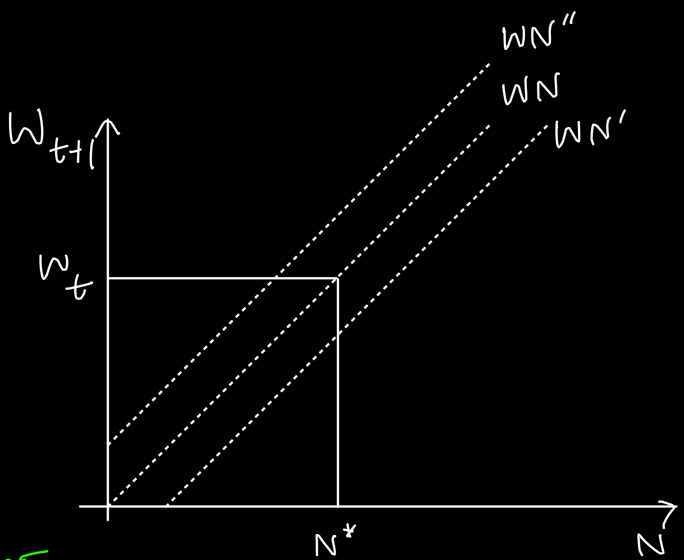
Phillip's Curve accounts for price stickness.

$$\boxed{g_w - \pi^e = -\epsilon(\underbrace{u - u^*}_{\text{unemployment gap}})}$$

$\frac{N^* - N}{N^*} \longrightarrow$ unemployment rate

$$u - u^* = \frac{N^* - N}{N^*}$$

$$\boxed{g_m - \pi^e = \frac{W_{t+1} - W_t}{W_t} - \pi^e = -\epsilon\left(\frac{N^* - N}{N^*}\right)}$$



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Wage - employment relationship is dynamic.

Empirically wage adjustments + prices are sluggish.

EDWARD* THEORY

Workers are often fooled into thinking that a nominal wage increase is beneficial whereas in reality the wage increase is below price inflation.

Reason for stickiness

Coordination Failure

① $M^S \uparrow \Longrightarrow P \uparrow$

However, in reality there is a delay while prices increase due to coordination failure among private firms.

Due to differing changes in demand for the firms' products.

Delay in price adjustment

* PRICE CEILING
* WAGE FLOOR : Wages are sticky downwards.

When AD is low (recession) wages take a hit

Last resort

$$\boxed{\frac{y-y^*}{y^*} = -w(u-u^*)}$$ OKUN'S LAW

If unemployment goes down GDP goes up

-ve correlation

Eg:- Say a unit production, W wage

a $\quad$ W

$$\frac{W}{a} = \text{cost per unit}$$

$$P = \frac{(1+z)W}{a}$$

z: markup

$$\hat{p} = \frac{dp}{dt} = \frac{dW}{dt} = \hat{W}$$

$$\frac{W_{t+1}}{W_t} - \pi^e = -\epsilon\left(\frac{N^*-N}{N^*}\right) \quad / \quad \frac{y-y^*}{y^*} = -w(u-u^*)$$

$\Downarrow$ wage and price inflation are interchangeable $\qquad w \simeq 2$

$$\frac{P_{t+1}}{P_t} - \pi^e = \frac{\epsilon}{w}\left(\frac{y-y^*}{y^*}\right)$$

$$P_{t+1}^e = P_t$$

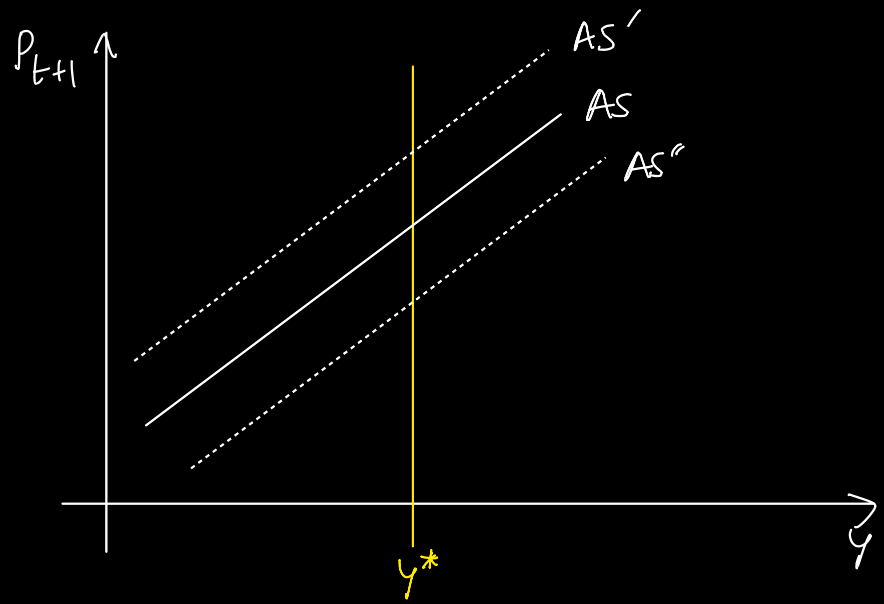
$$\boxed{P_{t+1} = P_{t+1}^e + P_t\frac{\epsilon}{w}\left(\frac{y-y^*}{y^*}\right)}$$

$$P_{t+1} = P_{t+1}^e\left(1+\lambda(y-y^*)\right)$$

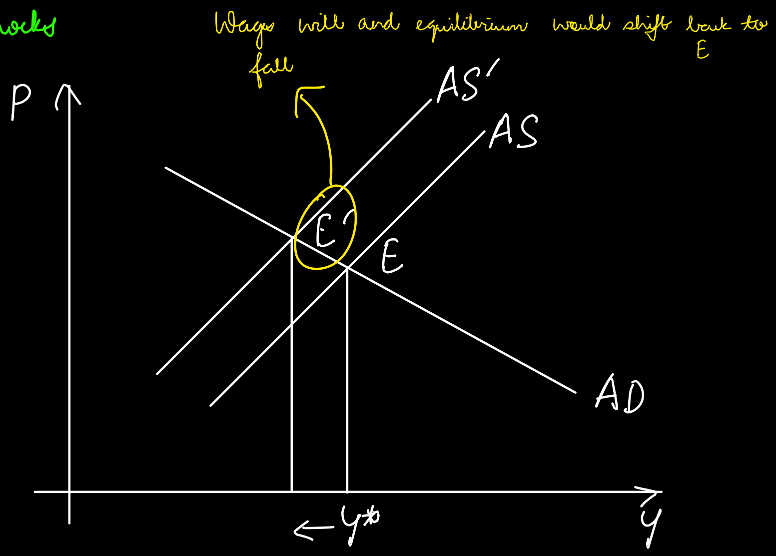
Efficiency Wages :- For highly valuable or productive workers wages are often higher than equilibrium wages.

Menu Cost : Cost to increasing prices, reduction in revenue due to losing customers to a price increase.

DERIVATION OF AGGREGATE SUPPLY CURVE



* Supply shocks



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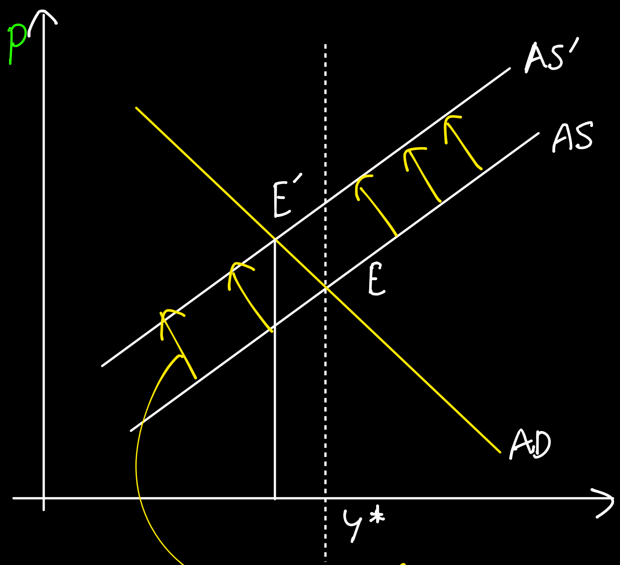
SHOCKS



Cost per unit increased due to oil price shock.

Oil Price Shock

Temporary unemployment → economy does eventually get back to original point of equilibrium E.

Prices are faster to adjust than wages

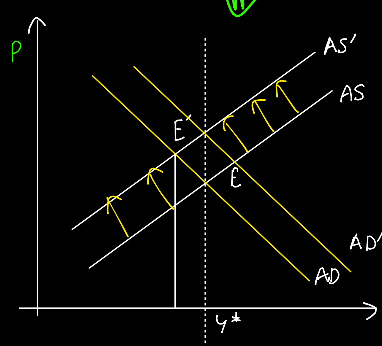
① Due to shock W↓, P↑

② Over time P goes back, P↓ but wages do not adjust so fast

Government would step in } accomodative policy

One could ask why this was not done early on

It is about balancing tradeoffs, you do not want to intervene early on and stifle output



SHOCKS → Temporary shocks
SHOCKS → Permanent shocks

Accomodation policies are effective only on temporary shocks

FAVOURABLE SHOCKS

Eg:- Massive reduction in compute power.

Policy makers must increase the demand being cautious so as to not increase price levels too much.